

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: CSA16 COURSE CODE: Data Warehousing and Data Mining

Q1. Supermarket Customer Segmentation Using Clustering

Clustering Approach:

K-Means clustering is used to segment supermarket customers based on their monthly visits, average spending, and number of items purchased. The Customer ID is excluded because it is only an identifier and should not affect the clustering process. In WEKA, the **SimpleKMeans** algorithm is applied with **K = 2**.

WEKA Result:

The dataset contains 5 customers and 3 clustering attributes. The algorithm formed two clusters.

- **Cluster 0:** C2 and C4
 - Average visits: 3.5/month
 - Average spending: ₹5,500
 - Average items bought: 7
- **Cluster 1:** C1, C3 and C5
 - Average visits: 11/month
 - Average spending: ₹1,966.67
 - Average items bought: 16.33

Customer Segmentation:

Cluster 0 represents customers who visit the supermarket less frequently but have high spending per visit. Cluster 1 represents frequent and highly engaged customers who purchase a larger number of items.

Targeted Marketing:

For Cluster 0, the supermarket can provide personalized coupons, special offers, and visit-frequency incentives to encourage more visits. For Cluster 1, loyalty programs, reward points, bundle offers, and personalized product recommendations can be provided to maintain their engagement and encourage repeat purchases.

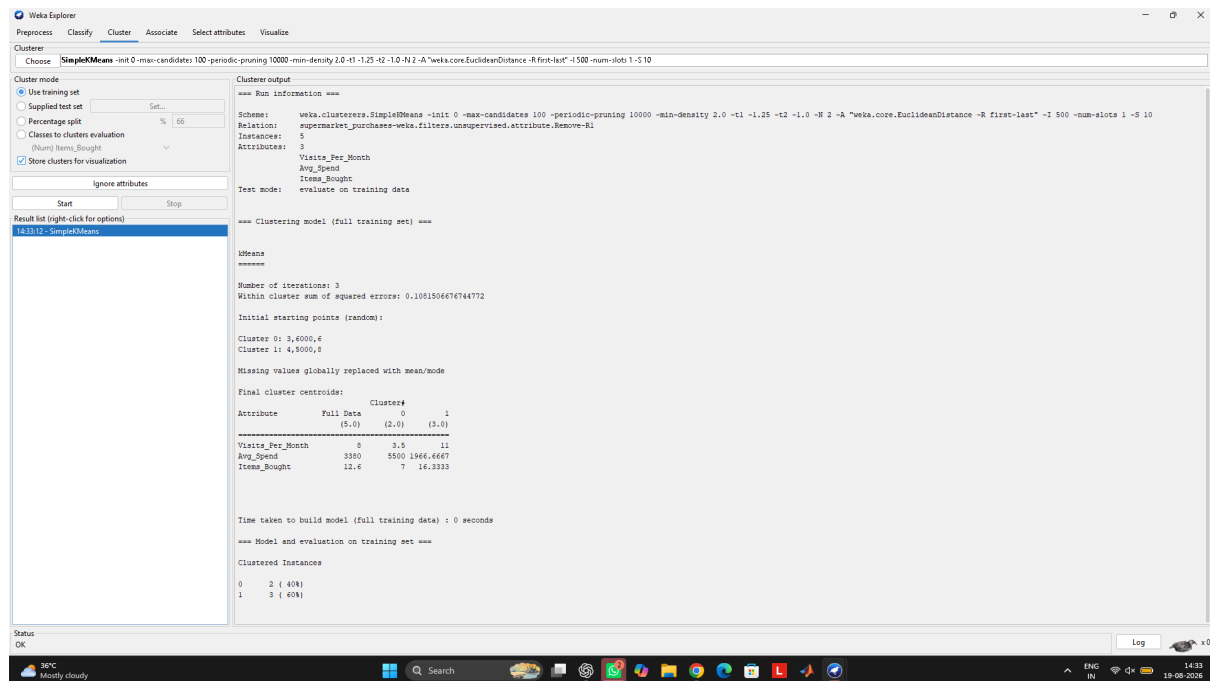
Conclusion:

K-Means clustering helps the supermarket identify different customer behaviour patterns. By understanding these segments, the business can design targeted marketing campaigns instead of using the same strategy for every customer. This can improve customer satisfaction, customer retention, and sales.

WEKA Tool Used: WEKA Explorer – SimpleKMeans

Number of Clusters: 2

Clustering Features: Visits/Month, Average Spend, Items Bought



Q2. Credit Card Fraud Detection Using Clustering

Clustering Approach:

A clustering-based anomaly detection approach can be used to identify unusual credit card transactions. The **SimpleKMeans** algorithm in WEKA is applied with **K = 2**. Transaction ID is removed because it is only an identifier. The transaction amount, location code, and transaction time are used as clustering features. Since the attributes have different scales, normalization is applied before clustering.

WEKA Result:

The dataset contains 5 transactions and 3 clustering attributes. SimpleKMeans produced two clusters.

- **Cluster 0:** T2 and T4
 - T2: ₹5,000, Delhi, 2 hrs
 - T4: ₹7,000, Mumbai, 1 hr
 - High transaction amounts and unusual transaction times/locations.
- **Cluster 1:** T1, T3 and T5
 - T1: ₹200, Chennai, 10 hrs
 - T3: ₹150, Chennai, 11 hrs
 - T5: ₹180, Chennai, 12 hrs
 - Low transaction amounts with a similar location and time pattern.

Fraud Detection:

Cluster 0 is considered the suspicious cluster because its transactions have substantially higher amounts and different location/time characteristics compared with the normal transaction cluster. Transactions T2 and T4 should therefore be flagged for additional fraud verification.

Practical Solution:

In a real credit card system, the clustering model can continuously monitor transactions. Transactions that are far from the normal cluster can be assigned a higher risk score. The system can then trigger additional verification, temporary transaction review, or customer confirmation before completing the transaction.

Analysis:

The WEKA model successfully separates the five transactions into two behavioural groups. Cluster 0 contains 2 transactions (40%), while Cluster 1 contains 3 transactions (60%). The clustering result demonstrates how unusual transaction behaviour can be separated from a normal pattern.

Conclusion:

K-Means clustering provides a useful unsupervised method for detecting unusual credit card transaction patterns. It can support fraud detection by identifying transactions that require further investigation without requiring pre-labelled fraudulent transaction data.

WEKA Tool: WEKA Explorer – SimpleKMeans

Number of Clusters: 2

Preprocessing: Transaction ID removed and numerical attributes normalized

The screenshot shows the WEKA Explorer interface with the SimpleKMeans clustering algorithm applied. The left pane shows the 'Cluster' tab with 'SimpleKMeans' selected. The right pane displays the 'Cluster output' window, which contains the following information:

```
=== Run information ===
Scheme: weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -E 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10
Relation: credit_card_transactions-weka.filters.unsupervised.attribute.Remove-RI-weka.filters.unsupervised.attribute.Normalize-S1.0-70.0
Instances: 5
Attributes: 3
Amount
Location_Code
Time_Hours
Test mode: evaluate on training data

=== Clustering model (full training set) ===

KMeans
=====
Number of iterations: 3
Within cluster sum of squared errors: 0.1893116267710961

Initial starting points (random):
Cluster 0: 1,1,1
Cluster 1: 0.708029,0.5,2

Missing values globally replaced with mean/mode

Final cluster centroids:
Attribute      Full Data      Cluster#
              (5,0)          (2,0)          (3,0)
-----
Amount          0.3439         0.594         0.0039
Location_Code    0.3           0.75          0
Time_Hours       7.2           1.5          11

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances
0      2 ( 40%)
1      3 ( 60%)
```

Q3. Telecom Customer Churn Identification Using Clustering

Clustering Approach:

Clustering can be used to divide telecom users into groups according to their usage behaviour. The **SimpleKMeans** algorithm is applied in WEKA with **K = 2**. The User ID is removed because it is only an identifier. Call duration, data usage, and recharge amount are used as clustering features. Normalization is applied so that attributes with different numerical scales do not dominate the clustering process.

WEKA Result:

The dataset contains 5 users and 3 clustering attributes. The K-Means algorithm produces two customer groups.

- **Cluster 0:** U2 and U4
 - Low call duration
 - Low data usage
 - Low recharge amount
- **Cluster 1:** U1, U3 and U5
 - High call duration
 - High data usage
 - Higher recharge amount

Churn Identification:

Cluster 0 can be considered the potentially churn-prone group because U2 and U4 have relatively low usage and recharge values. Low engagement may indicate that these users are less connected to the service and could be at greater risk of switching to another telecom provider.

Customer Retention Strategy:

The telecom company can target U2 and U4 with personalized low-cost recharge plans, additional data benefits, bonus talk time, discounts, and special retention offers. The company can also monitor changes in their usage behaviour over time to identify increasing churn risk.

Analysis:

The WEKA model divided the five users into two groups. Cluster 0 contains 2 users (40%), while Cluster 1 contains 3 users (60%). The clustering clearly separates low-usage users from highly active users, making it useful for customer segmentation and retention planning.

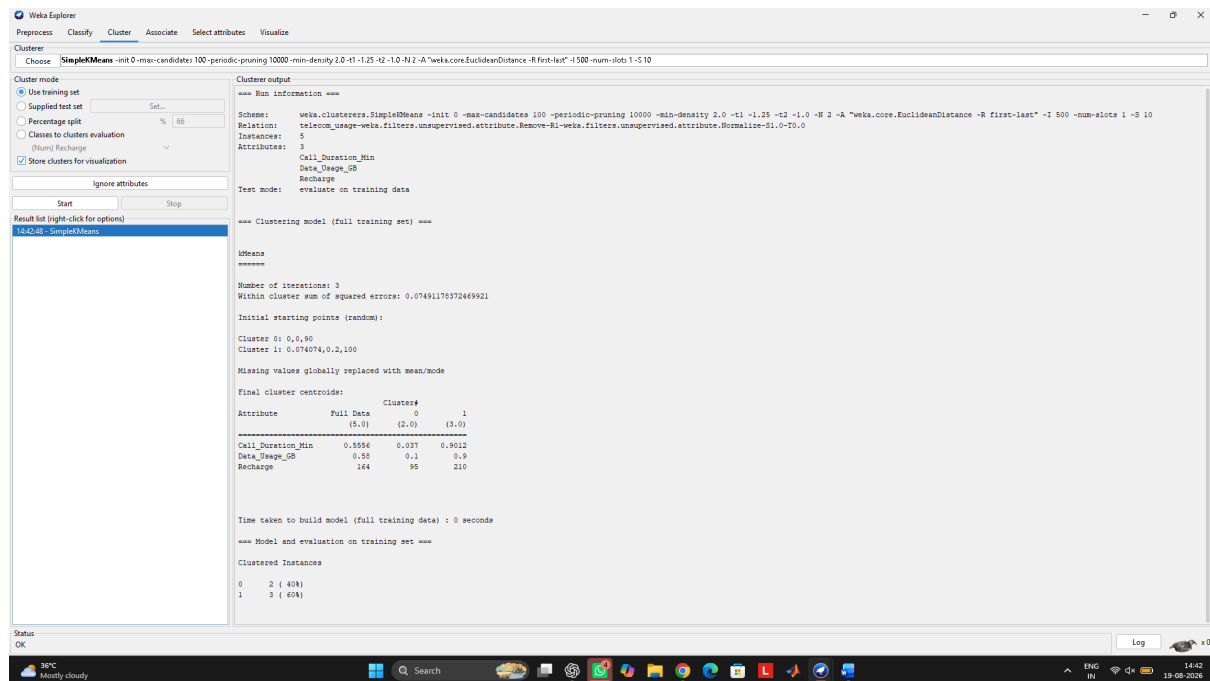
Conclusion:

Clustering provides an effective unsupervised method for identifying different telecom usage patterns. By detecting low-engagement users, telecom companies can apply targeted retention strategies and potentially reduce customer churn.

WEKA Tool: WEKA Explorer – SimpleKMeans

Number of Clusters: 2

Preprocessing: User ID removed and numerical attributes normalized



Q4. E-Commerce User Clustering for Recommendation Systems

Clustering Technique:

The **K-Means clustering algorithm** can be used to group e-commerce users according to their online behaviour. In WEKA, the **SimpleKMeans** algorithm is applied with **K = 2**. The User ID is removed because it is only an identifier. Product views, clicks, and purchases are used as clustering features. Normalization is applied to make the numerical attributes comparable.

WEKA Result:

The dataset contains 5 users and 3 clustering attributes. The algorithm produces two user groups.

- **Cluster 0:** U2 and U4
 - Low product views
 - Low clicks
 - Low purchases
- **Cluster 1:** U1, U3 and U5
 - High product views
 - High clicks
 - High purchases

User Segmentation:

Cluster 0 represents low-engagement users who interact less with products and make fewer purchases. Cluster 1 represents highly engaged users who view more products, click more frequently, and make more purchases.

Recommendation System Improvement:

The recommendation system can use these clusters to provide personalized recommendations. For

Cluster 0, the system can recommend popular products, discounted products, similar products, and personalized offers to increase engagement. For Cluster 1, the system can recommend complementary products, related products, premium products, and frequently purchased items.

Analysis:

The WEKA model divided the five users into two groups. Cluster 0 contains 2 users (40%), while Cluster 1 contains 3 users (60%). The clustering separates low-engagement users from highly engaged users based on their browsing and purchasing behaviour.

Conclusion:

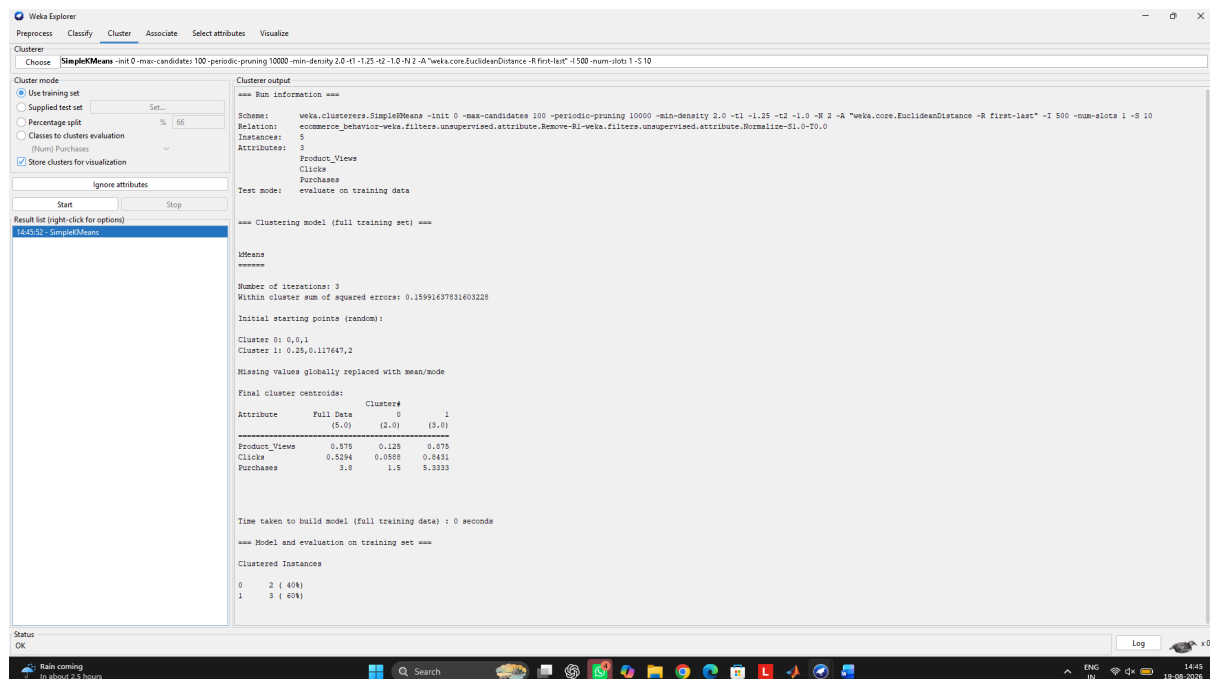
Clustering improves recommendation systems by grouping users with similar behaviour. Instead of giving the same recommendations to every customer, the e-commerce platform can provide cluster-specific recommendations. This can improve personalization, user engagement, and purchase opportunities.

WEKA Tool: WEKA Explorer – SimpleKMeans

Number of Clusters: 2

Clustering Features: Product Views, Clicks, Purchases

Preprocessing: User ID removed and numerical attributes normalized



Q5. Traffic Congestion Zone Identification Using Clustering

Clustering Approach:

A clustering-based approach can be used to identify traffic congestion zones by grouping locations with similar traffic characteristics. The **SimpleKMeans** algorithm is applied in WEKA with **K = 2**. The Location ID is removed because it is only an identifier. Vehicle count, average speed, and delay are used as clustering features. Normalization is applied so that the attributes are on comparable scales.

WEKA Result:

The dataset contains 5 locations and 3 clustering attributes. The K-Means algorithm produces two clusters.

- **Cluster 0:** L2 and L4
 - High vehicle count
 - Low average speed
 - High delay
- **Cluster 1:** L1, L3 and L5
 - Lower vehicle count
 - Higher average speed
 - Lower delay

Congestion Identification:

Cluster 0 is identified as the high-congestion group. L2 has 500 vehicles, an average speed of 15 km/h, and a delay of 25 minutes. L4 has 600 vehicles, an average speed of 10 km/h, and a delay of 30 minutes. These characteristics indicate severe traffic congestion.

Cluster 1 represents relatively low-congestion locations because the vehicle counts are lower, average speeds are higher, and delays are smaller.

Traffic Management Strategy:

For high-congestion locations such as L2 and L4, traffic authorities can optimize traffic signal timings, deploy traffic personnel during peak periods, suggest alternate routes, and use intelligent traffic management systems. Real-time sensors and cameras can also be used to continuously monitor vehicle density and congestion.

Analysis:

The WEKA model divided the five locations into two groups. Cluster 0 contains 2 locations (40%), while Cluster 1 contains 3 locations (60%). The clustering successfully separates high-congestion areas from relatively free-flowing areas based on vehicle count, speed, and delay.

Conclusion:

K-Means clustering can help traffic authorities identify congestion zones automatically from traffic data. The identified high-congestion zones can be prioritized for signal optimization, traffic control, route planning, and infrastructure improvements.

WEKA Tool: WEKA Explorer – SimpleKMeans

Number of Clusters: 2

Clustering Features: Vehicle Count, Average Speed, Delay

Preprocessing: Location ID removed and numerical attributes normalized

Weka Explorer

PreprocessClassifyClusterAssociateSelect attributesVisualize

Cluster

ChooseSimpleKMeans-init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Cluster mode

Use training set

Supplied test set

Percentage split

Classes to clusters evaluation

When Delay_Min

Store clusters for visualization

Ignore attributes

Start

Stop

Result list (right-click for options)

143007 - SimpleKMeans

Cluster output

=== Run information ===

Scheme: weka.clusterers.SimpleKMeans -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 2 -A "weka.core.EuclideanDistance -R first-last" -I 500 -num-slots 1 -S 10

Relation: traffic_data-weka.filters.unsupervised.attribute.Remove-R1-weka.filters.unsupervised.attribute.Normalize-R1.0-70.0

Instances: 5

Attributes: 3

Vehicle_Count

Avg_Speed

Delay_Min

Test mode: evaluate on training data

=== Clustering model (full training set) ===

KMeans

====

Number of iterations: 3

Within cluster sum of squared errors: 0.1137620536465107

Initial starting points (random):

Cluster 0: 1,0,30

Cluster 1: 0.75,0.227273,25

Missing values globally replaced with mean/mode

Final cluster centroids:

Attribute	Full Data	Clusters	
		0	1
	(5.0)	(2.0)	(5.0)
Vehicle_Count	0.345	0.875	0.025
Avg_Speed	0.5909	0.1134	0.9091
Delay_Min	27.2	27.5	10.3333

Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0	2 (40%)
1	3 (60%)

Status

OK

Log

x 0

36°C

Mostly cloudy

Search

ENG

IN

14:50

19-09-2025